PES UNIVERSITY, BANGALORE

Department of Computer Science and Engineering

UE24CS341A — Software Engineering

Software Requirements Specification

for

Student Attendance Management System

Version 1.1 (Pending Approval)

Updated to reflect the work done in Sprints 1 and 2

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Revision History

Version	Date	Author	Change Summary	Approval
1.0	03-09-2026	Team 3	Initial SRS document with system requirements, use-case diagrams, and RTM	Pending
1.1	30-10-2026	Team 3	Updated to reflect the work done in Sprints 1 and 2: time-slot wording (SAM-F-012, SAM-F-013, UC-01), RTM status, Section 10	Pending

Approvals

Role	Name	Signature / Email	Date
Course Coordinator			

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1. Introduction

1.1 Purpose

This document is the Software Requirements Specification (SRS) for the Student Attendance Management System (SAMS), developed by Team 3 as part of the UE24CS341A Software Engineering mini-project. It defines the functional and non-functional requirements, system interfaces, features, quality attributes, and verification criteria of the system. The document serves as the reference baseline for developers, faculty members, administrators, students, and project evaluators through the requirements, design, implementation, and testing phases of the project.

1.2 Scope

The Student Attendance Management System automates the recording, management, monitoring, and reporting of student attendance, replacing manual attendance registers with a centralized digital platform. Faculty members mark attendance for their assigned classes, students view their own attendance records, and administrators manage students, faculty, subjects, and system-wide attendance data.

Core capabilities include user authentication with role-based access, attendance recording and correction, automatic attendance-percentage calculation, attendance history and reporting, and low-attendance alerts.

Out of scope: payroll management, examination evaluation, fee management, and any other non-academic administrative activity not directly related to attendance tracking.

1.3 Intended Audience

- Developers, who need to understand and implement the system requirements.
- Faculty members, who will record and monitor student attendance.
- Students, who will view their attendance status and history.
- Administrators, who manage users, subjects, classes, and system data.
- QA/test engineers, who verify and validate system functionality against this SRS.
- Project guides and evaluators, who assess requirement coverage and implementation quality.

The remainder of this document is organized following the standard SRS structure: overall system description, external interfaces, detailed functional and non-functional requirements, quality attributes, use-case diagrams, and the requirements traceability matrix.

1.4 Definitions and Acronyms

Term / Acronym	Definition
SRS	Software Requirements Specification
SAM	Student Attendance Management
UI	User Interface
API	Application Programming Interface
DB	Database
Admin	System Administrator
Faculty	Teacher responsible for marking attendance
Student	User whose attendance is recorded

Term / Acronym	Definition
Attendance	Record indicating whether a student was present or absent
CRUD	Create, Read, Update, Delete
ID	Identification Number
Authentication	Process of verifying the identity of a user
Authorization	Process of determining what a user is permitted to access

1.5 References

- IEEE/software engineering SRS guidelines, as adapted in the Mini-Project SRS template (PES University, Dept. of CSE).
- UML modelling standards for use-case diagrams.
- Django Software Foundation, "Django Documentation", https://docs.djangoproject.com/
- Oracle Corporation, "MySQL 8.0 Reference Manual", https://dev.mysql.com/doc/
- Institutional attendance rules and academic requirements (as communicated by the course coordinator).

2. Overall Description

2.1 Product Perspective

The Student Attendance Management System (SAMS) is a new, self-contained web-based application; it is not a follow-on to an existing institutional system. It replaces manual, register-based attendance tracking with a centralized system comprising a browser-based user interface, an application server, and a relational database.

The system uses Python with the Django web framework for the application layer and MySQL for relational persistence. Django was selected for its built-in authentication module, role-based authorization support, and ORM capabilities, which streamline entity mapping for students, faculty, courses, and attendance records. MySQL provides standard relational integrity, acid-compliant transactions, and native foreign-key constraints to prevent data duplication.

Faculty members record attendance for their assigned subjects, students view their own attendance status, and administrators manage students, faculty, subjects, classes, and attendance records. A high-level view of the major components and their interconnections is provided as a structural note: the browser-based client communicates over HTTPS with the Django application server, which in turn performs all reads and writes against the MySQL database; no other external system integration is required for the current scope.

2.2 Product Functions

The major functions of the system, summarized at a high level (details are in Section 4), are:

- User Authentication: secure login with role-based access for Admin, Faculty, and Student.
- Student Management: add, update, search, and view student records.
- Faculty and Subject Management: maintain faculty profiles, create subjects, and assign subjects to faculty.
- Attendance Recording: mark students Present/Absent per class and subject, with duplicate-entry prevention.
- Attendance Modification: authorized correction of previously recorded attendance.
- Attendance Monitoring: automatic percentage calculation and history viewing.
- Attendance Reports: student-wise, subject-wise, and class-wise reports, downloadable/printable.
- Low-Attendance Alerts: automatic warnings when a student's attendance falls below the configured threshold.
- Dashboards: role-specific dashboards for Admin, Faculty, and Student.

2.3 User Classes and Characteristics

Student

- Logs in using valid credentials issued by the institution.
- Views personal attendance records, subject-wise percentage, and history.
- Receives low-attendance warnings.
- Requires only basic computer or smartphone familiarity; this is the most frequent but least privileged user class.

Faculty

- Logs in securely and views assigned classes and subjects.
- Marks and corrects attendance, and generates attendance reports for their subjects.
- Requires basic familiarity with the institution's attendance process; moderate frequency of use, moderate privilege.

Administrator

- Has full administrative access: manages student and faculty accounts, departments, classes, and subjects.
- Assigns subjects to faculty and views consolidated, institution-wide attendance reports.
- Least frequent user class but highest privilege level; most important for data integrity and system configuration.

2.4 Operating Environment

- Client: desktop, laptop, tablet, or smartphone with a modern web browser (Chrome, Edge, Firefox).
- Frontend: HTML, CSS, JavaScript, with Bootstrap for responsive layout.
- Backend: Python 3 with the Django web framework.
- Database: MySQL (relational).
- Server operating system: Linux (Ubuntu), compatible with Windows/macOS for local development.
- Network: institutional LAN or the internet, secured with HTTPS.

2.5 Design and Implementation Constraints

- The backend framework and database are fixed as Django and MySQL for this project (Section 2.1); no alternate stack will be introduced mid-project.
- Only authenticated users may access protected system functions, and access must match the user's assigned role.
- Only authorized faculty or administrators may modify attendance records once submitted.
- Passwords must be stored using a one-way hashing algorithm (Django's default PBKDF2/Argon2 password hasher); plaintext storage is prohibited.
- Regular database backups must be scheduled to prevent loss of attendance records.
- The system must follow the attendance rules and policies defined by the institution (e.g., minimum attendance percentage thresholds).

2.6 Assumptions and Dependencies

- It is assumed that all students and faculty are issued institutional credentials prior to system rollout; the system does not perform its own identity verification against external ID documents.
- The system depends on continuous availability of the MySQL database and the underlying network for normal operation; extended outages will degrade functionality.
- It is assumed that class timetables and subject-to-semester mappings are supplied by the administrator and are accurate at the time attendance is recorded.
- The project depends on Django's built-in authentication and ORM libraries, which are assumed to remain stable for the duration of development.

3. External Interface Requirements

3.1 User Interfaces

The system provides a simple, role-aware web interface:

- A single login page shared by students, faculty, and administrators, differentiated by role after authentication.
- A role-specific dashboard shown immediately after login (Section 2.2).
- Forms for entering and updating student, faculty, subject, and attendance details, with inline validation.
- An attendance-marking interface presenting each enrolled student with a Present/Absent toggle.
- Attendance-percentage and report views with clear tabular layouts.
- Consistent navigation menus, clear primary-action buttons, and legible error messages.
- A responsive layout usable on desktop, tablet, and smartphone screens.

3.2 Hardware Interfaces

The system requires no specialized hardware and runs on standard computing devices.

- Keyboard and mouse, or touchscreen, for data entry and navigation.
- A standard display for viewing the application.
- A printer, optionally, for physical copies of attendance reports.
- Standard server hardware (or a cloud virtual machine) to host the Django application and MySQL database.

3.3 Software Interfaces

- Database: MySQL, accessed through the Django ORM, for storing student, faculty, subject, and attendance data.
- Backend framework: Django (Python 3), handling request routing, business logic, and template rendering.
- Web browser: Google Chrome, Microsoft Edge, or Mozilla Firefox (current stable releases).
- Authentication module: Django's built-in authentication system, extended with role-based permission checks for Admin, Faculty, and Student.
- Reporting module: a Django-based reporting component that queries attendance data and renders it as on-screen tables and downloadable/printable documents.

3.4 Communications Interfaces

- All communication between the browser, application server, and database occurs over the institutional network or the internet.
- HTTPS (TLS) is used to protect all data transmitted between the client and the server.
- The system displays clear, user-facing error messages when the network or server is unavailable, rather than failing silently.
- Database connections are restricted to the application server and secured to prevent unauthorized direct access.

4. System Features (Detailed)

Each functional requirement below includes an acceptance criterion and a reference test case. Requirement IDs follow the pattern SAM-F-###. A total of 20 functional requirements are specified across six functional domains.

4.1 User Authentication

Description: The system shall authenticate students, faculty members, and administrators before granting access, and shall enforce role-based access thereafter.

Req ID	Requirement	Type	Priority	Source / Stakeholder	Acceptance Criteria / Test Case Ref	Comments / Dependencies
SAM-F-001	The system shall allow users to log in using a valid username and password.	Functional	High	All Users	Valid credentials shall provide access. Test: TC-Auth-01	Requires user database
SAM-F-002	The system shall display an appropriate error message when invalid login credentials are entered.	Functional	High	Security	Invalid credentials shall be rejected. Test: TC-Auth-02	Depends on authentication module
SAM-F-003	The system shall provide role-based access for Admin, Faculty, and Student users.	Functional	High	Administrator	Users shall access only features permitted for their role. Test: TC-Auth-03	Requires role management

4.2 Student Management

Description: The system shall allow administrators to maintain student information.

Req ID	Requirement	Type	Priority	Source / Stakeholder	Acceptance Criteria / Test Case Ref	Comments / Dependencies
SAM-F-004	The system shall allow the administrator to add new student details.	Functional	High	Administrator	New student shall be successfully stored. Test: TC-Student-01	Requires database
SAM-F-005	The system shall allow the administrator to update student information.	Functional	High	Administrator	Modified details shall be saved correctly. Test: TC-Student-02	Admin authorization required
SAM-F-006	The system shall allow the administrator to search and view	Functional	Medium	Administrator/ Faculty	Correct student information shall be displayed. Test:	Search by student ID/name

Req ID	Requirement	Type	Priority	Source / Stakeholder	Acceptance Criteria / Test Case Ref	Comments / Dependencies
	student details.				TC-Student-03	

4.3 Faculty and Subject Management

Description: The system shall allow the administrator to manage faculty members, subjects, and class assignments.

Req ID	Requirement	Type	Priority	Source / Stakeholder	Acceptance Criteria / Test Case Ref	Comments / Dependencies
SAM-F-007	The system shall allow the administrator to add and manage faculty details.	Functional	High	Administrator	Faculty information shall be stored and displayed correctly. Test: TC-Faculty-01	Requires database
SAM-F-008	The system shall allow the administrator to create and manage subjects.	Functional	High	Administrator	Subject details shall be successfully added or updated. Test: TC-Subject-01	Requires subject database
SAM-F-009	The system shall allow the administrator to assign subjects to faculty members.	Functional	High	Administrator	Assigned subject shall appear on the faculty dashboard. Test: TC-Subject-02	Faculty account required

4.4 Attendance Recording

Description: Faculty members shall be able to record attendance for students belonging to their assigned classes.

Req ID	Requirement	Type	Priority	Source / Stakeholder	Acceptance Criteria / Test Case Ref	Comments / Dependencies
SAM-F-010	The system shall allow faculty members to select a class and subject before marking attendance.	Functional	High	Faculty	Correct class and subject shall be displayed. Test: TC-Att-01	Requires class/subject data

SAM-F-011	The system shall allow faculty members to mark each student as Present or Absent.	Functional	High	Faculty	Attendance status shall be stored correctly. Test: TC-Att-02	Requires student list

Req ID	Requirement	Type	Priority	Source / Stakeholder	Acceptance Criteria / Test Case Ref	Comments / Dependencies
SAM-F-012	The system shall store attendance with student ID, subject, date, time slot, and attendance status.	Functional	High	Faculty/ Administrator	Database record shall contain all required fields. Test: TC-Att-03	Requires database
SAM-F-013	The system shall prevent duplicate attendance records for the same student, subject, date, and time slot.	Functional	Medium	System	Duplicate record shall be rejected. Test: TC-Att-04	Requires database constraint

4.5 Attendance Monitoring and Modification

Req ID	Requirement	Type	Priority	Source / Stakeholder	Acceptance Criteria / Test Case Ref	Comments / Dependencies
SAM-F-014	The system shall allow authorized faculty members to modify an attendance record.	Functional	High	Faculty	Authorized changes shall be saved successfully. Test: TC-Att-05	Authorization required
SAM-F-015	The system shall calculate the attendance percentage for each student and subject.	Functional	Medium	Student/Faculty	Percentage shall be calculated correctly. Test: TC-Calc-01	Based on attendance records
SAM-F-016	The system shall allow students to view their attendance history.	Functional	High	Student	Student shall see only their own attendance. Test: TC-View-01	Authentication required

SAM-F-017	The system shall display a warning when a student's attendance falls below the required percentage.	Functional	Medium	Student/Faculty	Low-attendance student shall receive a warning. Test: TC-Alert-01	Attendance threshold configurable

4.6 Attendance Reports

Req ID	Requirement	Type	Priority	Source / Stakeholder	Acceptance Criteria / Test Case Ref	Comments / Dependencies
SAM-F-018	The system shall generate student-wise attendance reports.	Functional	High	Faculty/Admin	Report shall display correct attendance details. Test: TC-Report-01	Requires attendance data
SAM-F-019	The system shall generate subject-wise and class-wise attendance reports.	Functional	High	Faculty/Admin	Selected subject/class report shall be generated. Test: TC-Report-02	Requires reporting module
SAM-F-020	The system shall allow authorized users to download or print attendance reports.	Functional	Medium	Faculty/Admin	Report shall be available in printable/downloadable format. Test: TC-Report-03	Requires report generation

Total Functional Requirements: 20

5. Non-Functional Requirements (Detailed)

IDs follow the pattern SAM-NF-###. The following non-functional requirements define the performance, reliability, usability, and maintainability criteria for the system.

Req ID	Requirement	Category	Priority	Acceptance Criteria / Measurement
SAM-NF-001	The system shall display normal user requests within 3 seconds under normal load.	Performance	High	90% of requests complete within 3 seconds. Test: TC-Perf-01
SAM-NF-002	The system shall provide at least 99% availability during academic working hours.	Reliability	High	Monthly availability report shall meet the target. Test: TC-Reliability-01
SAM-NF-003	The system shall provide a simple and user-friendly interface for students, faculty, and administrators.	Usability	High	Representative users shall complete common tasks without assistance. Test: TC-UX-01
SAM-NF-004	The system shall protect stored attendance and user information from unauthorized access.	Security	High	Unauthorized access attempts shall be denied. Test: TC-Sec-01
SAM-NF-005	The system shall support regular database backup and recovery of attendance records.	Reliability	High	Backup and restoration shall successfully recover test data. Test: TC-Recovery-01
SAM-NF-006	The system shall be maintainable so that new features can be added without major changes to existing modules.	Maintainability	Medium	New module integration shall not break existing functions. Test: TC-Maint-01
SAM-NF-007	The system shall support multiple users accessing the system simultaneously without significant performance degradation.	Scalability	Medium	Load testing shall meet defined response-time limits. Test: TC-Load-01

Total Non-Functional Requirements: 7

5.1 Security

5.1.1 Security Objectives

The Student Attendance Management System shall satisfy the following security objectives:

SO-01: Confidentiality. The system shall ensure that student personal information, login credentials, and attendance records are accessible only to authorized users.

SO-02: Integrity. The system shall ensure that attendance records cannot be modified, deleted, or manipulated by unauthorized users, and that stored attendance information remains accurate.

5.1.2 Security Requirements

Req ID	Security Requirement	Priority	Acceptance Criteria / Test
SAM-SEC-001	The system shall require authentication before allowing access to protected features.	High	Unauthenticated users shall be redirected to the login page. Test: TC-SEC-01
SAM-SEC-002	The system shall implement role-based access control for Admin, Faculty, and Student users.	High	Users cannot access functions outside their assigned role. Test: TC-SEC-02
SAM-SEC-003	Passwords shall be stored securely using a one-way password hashing mechanism and shall never be stored as plain text.	High	Database inspection shall confirm passwords are hashed. Test: TC-SEC-03
SAM-SEC-004	The system shall prevent students from viewing or modifying another student's attendance records.	High	Attempted unauthorized access shall be rejected. Test: TC-SEC-04
SAM-SEC-005	The system shall record important attendance modifications and administrative actions in an audit log.	Medium	Changes shall include user, action, and timestamp. Test: TC-SEC-05

Total Security Requirements: 5

6. Quality Attributes & Acceptance Tests

6.1 Quality Attributes

The Student Attendance Management System shall exhibit the following quality attributes:

- Usability: the system shall be simple and easy to use for students, faculty, and administrators.
- Performance: the system shall respond quickly to user requests (Section 5, SAM-NF-001).
- Reliability: attendance records shall be stored accurately without data loss.
- Security: only authorized users shall be able to access or modify attendance information.
- Maintainability: the system shall be easy to update and extend as new features are required.
- Scalability: the system shall support an increasing number of students and attendance records without significant degradation.

6.2 Acceptance Criteria

A build is considered acceptance-ready when: all High-priority functional requirements (Section 4) are implemented and pass their reference test cases; no critical non-functional requirement (Section 5) fails; all five security requirements (Section 5.1.2) pass; and the Requirements Traceability Matrix (Section 8) shows every requirement mapped to a design module and test case with no unresolved gaps.

6.3 Acceptance Test Cases

The following test suites shall be used to verify the system prior to final submission:

Test Suite	Purpose
Authentication Test	Verify login, logout, and role-based access.
Student Management Test	Verify adding, updating, searching, and viewing students.
Attendance Test	Verify marking, updating, and storing attendance.
Attendance Calculation Test	Verify correct attendance percentage calculation.
Report Test	Verify generation and downloading of attendance reports.
Performance Test	Verify that the system responds within the specified time.
Security Test	Verify authentication, authorization, password protection, and data access controls.

7. System Models and Diagrams

7.1 Actors

The Student Attendance Management System defines the following actors:

- Student: a user who views their own attendance records, percentage, and history.
- Faculty: a user who records and modifies attendance for their assigned classes and subjects, and generates reports.
- Administrator: a user who manages student, faculty, and subject data, assigns subjects to faculty, configures the attendance threshold, and views consolidated reports.
- Audit Module: a system-internal actor that receives and stores the log of sensitive administrative actions (Section 5.1.2, SAM-SEC-005).

7.2 Use Cases

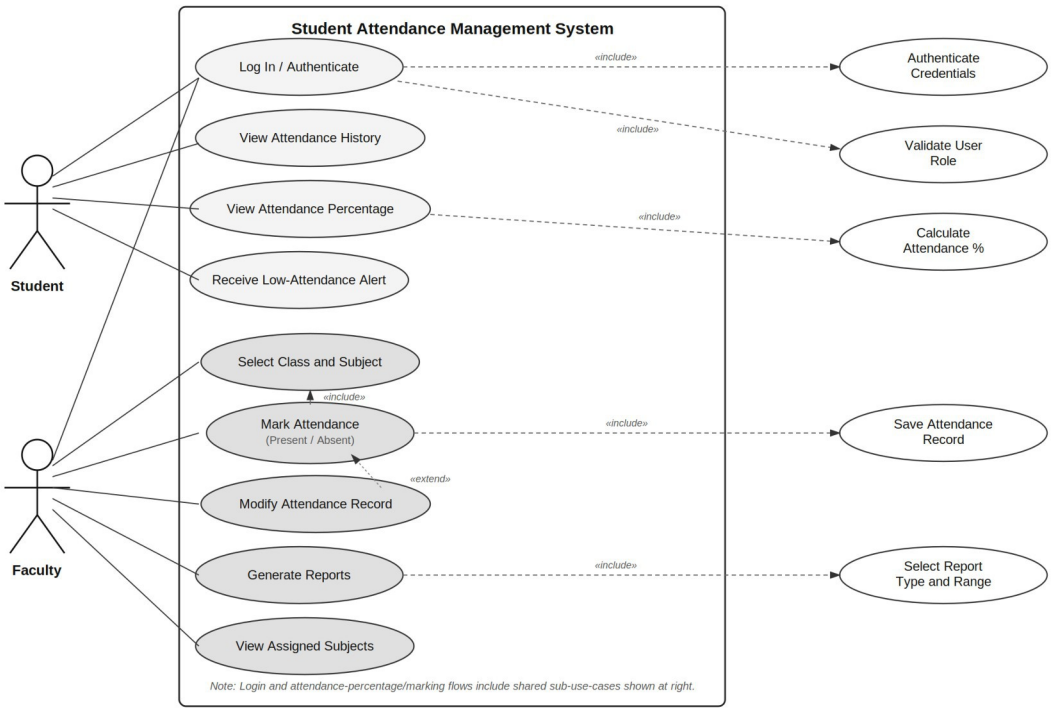
The primary use cases, grouped by actor, are summarized below and shown in the two use-case diagrams that follow (Section 7.3). Several use cases include a shared sub-use-case, shown as a separate oval connected by an «include» relationship: for example, Log In / Authenticate includes both Authenticate Credentials and Validate User Role, and Mark Attendance includes Select Class and Subject and Save Attendance Record.

- Student: Log In / Authenticate, View Attendance History, View Attendance Percentage, Receive Low-Attendance Alert.
- Faculty: Log In / Authenticate, View Assigned Subjects, Select Class and Subject, Mark Attendance, Modify Attendance Record, Generate Reports.
- Administrator: Manage Student Records, Manage Faculty Records, Manage Subjects and Classes, Assign Subjects to Faculty, Manage User Access, View Consolidated Attendance Reports, Configure Attendance Threshold, Manage Database Backup.

7.3 Use-Case Diagram

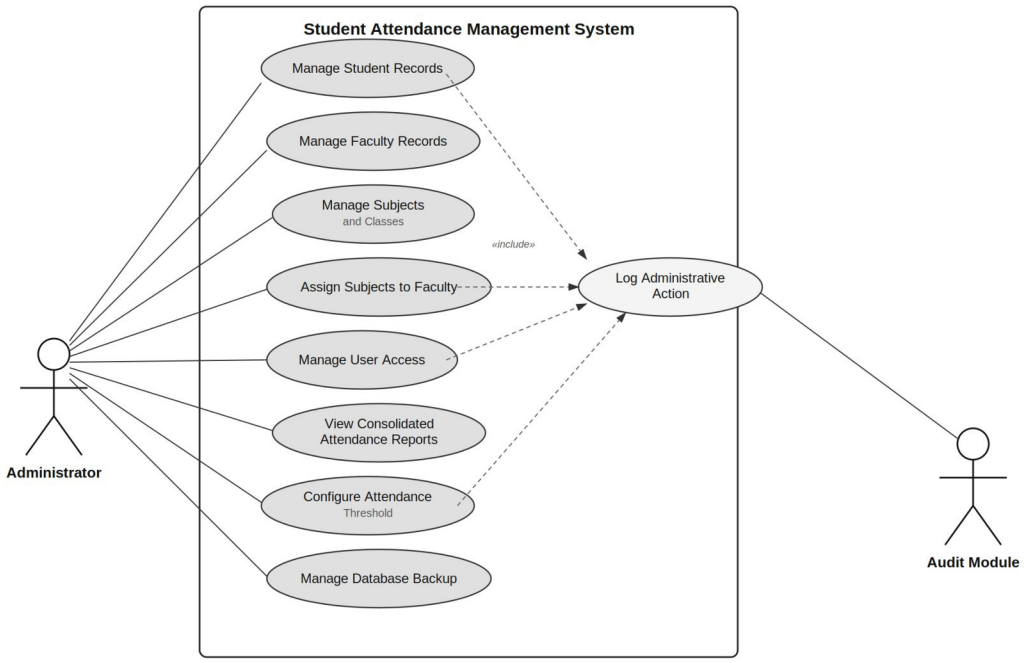
7.3.1 Student and Faculty Use Cases

The diagram below shows the Student and Faculty actors and their associated use cases. Login is shared by both actors and includes two sub-use-cases (Authenticate Credentials and Validate User Role); marking attendance includes selecting a class and subject beforehand and saving the resulting record; viewing attendance percentage includes the underlying percentage calculation; and generating reports includes selecting the report type and date range.



7.3.2 Administrator Use Cases

The diagram below shows the Administrator actor and the administrative use cases. Record-modifying and access-control actions (Manage Student Records, Assign Subjects to Faculty, Manage User Access, and Configure Attendance Threshold) each include Log Administrative Action, which writes to the Audit Module, in line with SAM-SEC-005; the same relationship applies to Manage Database Backup.



Note: the four record-modifying/access-control actions shown above include Log Administrative Action; the same relationship applies to Manage Database Backup (SAM-SEC-005).

7.4 Use-Case Descriptions

UC-01: Mark Attendance

Actor: Faculty. Precondition: Faculty is authenticated and has at least one assigned class/subject. Main flow: (1) Faculty selects class and subject (includes UC-02); (2) system displays the enrolled (active) student list; (3) Faculty picks the date and one or more one-hour time slots (two for a two-hour lab) and marks each student Present or Absent; (4) system stores one record per student per time slot with student ID, subject, date, time slot, and status in a single transaction, and saves nothing if any chosen slot is already marked for that date (SAM-F-012, SAM-F-013). Alternative flow: the faculty member later corrects a saved session, giving a reason that is kept in the audit log (SAM-F-014). Postcondition: attendance for the session is recorded and available for percentage calculation.

UC-02: Select Class and Subject

Actor: Faculty. Included by UC-01. Main flow: Faculty selects one of their assigned classes and subjects from a list scoped to their account; the system validates the selection before proceeding.

UC-03: View Attendance Percentage

Actor: Student. Precondition: Student is authenticated. Main flow: Student requests their attendance summary; the system calculates the percentage per subject from stored records (SAM-F-015) and displays it, along with a low-attendance warning if the percentage is below the configured threshold (SAM-F-017).

UC-04: Manage User Access

Actor: Administrator. Main flow: Administrator creates, updates, or deactivates a student or faculty account and assigns its role. This action includes UC-05 (Log Administrative Action) so that the change is auditable.

UC-05: Log Administrative Action

Actor: Audit Module (system). Triggered by: UC-04 and other record-modifying or access-control administrator actions (Manage Student Records, Assign Subjects to Faculty, Configure Attendance Threshold, Manage Database Backup). Main flow: the system records the acting user, the action taken, and a timestamp to the audit log (SAM-SEC-005).

UC-06: Generate Reports

Actor: Faculty or Administrator. Precondition: user is authenticated and has report-viewing permission for the requested scope. Main flow: user selects the report type (student-wise, subject-wise, or class-wise) and a date range (includes Select Report Type and Range); the system compiles the report from stored attendance records (SAM-F-018 to SAM-F-020) and returns it in an on-screen, printable, or downloadable format.

UC-07: View Attendance History

Actor: Student or Faculty. Precondition: user is authenticated. Main flow: user requests attendance history for a subject or class; the system retrieves matching records and displays them chronologically. Students see only their own records (SAM-SEC-004); faculty see records for classes they are assigned to.

8. Requirements Traceability Matrix (RTM)

The following RTM maps every requirement of the Student Attendance Management System to its SRS section, design module, and corresponding test case. Status: N = Not Tested, P = Pass, A = Approved. In version 1.1 every requirement has been verified by its test cases, which run automatically on every pull request in the GitHub Actions pipeline (STP v1.1, Appendix A); all are marked P, and move to A after the final review.

Req ID	Requirement (Short)	Section Ref / Design Spec	Module	Test Case(s)	Status	Comments
SAM-F-001	User login	4.1 / DS-Auth-01	Authentication Module	TC-Auth-01	P	—
SAM-F-002	Invalid login handling	4.1 / DS-Auth-02	Authentication Module	TC-Auth-02	P	—
SAM-F-003	Role-based access	4.1 / DS-Auth-03	Authentication Module	TC-Auth-03	P	—
SAM-F-004	Add student	4.2 / DS-Student-01	Student Module	TC-Student-01	P	—
SAM-F-005	Update student	4.2 / DS-Student-02	Student Module	TC-Student-02	P	—
SAM-F-006	Search student	4.2 / DS-Student-03	Student Module	TC-Student-03	P	—
SAM-F-007	Manage faculty	4.3 / DS-Faculty-01	Faculty Module	TC-Faculty-01	P	—
SAM-F-008	Manage subjects	4.3 / DS-Subject-01	Subject Module	TC-Subject-01	P	—
SAM-F-009	Assign subject to faculty	4.3 / DS-Subject-02	Subject Module	TC-Subject-02	P	—
SAM-F-010	Select class and subject	4.4 / DS-Attendance-01	Attendance Module	TC-Att-01	P	—
SAM-F-011	Mark attendance	4.4 / DS-Attendance-02	Attendance Module	TC-Att-02	P	—
SAM-F-012	Store attendance record	4.4 / DS-Attendance-03	Attendance Module	TC-Att-03	P	—
SAM-F-013	Prevent duplicate attendance	4.4 / DS-Attendance-04	Attendance Module	TC-Att-04	P	Database constraint
SAM-F-014	Modify attendance	4.5 / DS-Attendance-05	Attendance Module	TC-Att-05	P	Authorized users only
SAM-F-015	Calculate attendance percentage	4.5 / DS-Calc-01	Calculation Module	TC-Calc-01	P	—
SAM-F-016	View attendance history	4.5 / DS-View-01	Student Module	TC-View-01	P	—
SAM-F-017	Low-attendance warning	4.5 / DS-Alert-01	Notification Module	TC-Alert-01	P	Threshold configurable
SAM-F-018	Student-wise report	4.6 / DS-Report-01	Report Module	TC-Report-01	P	—
SAM-F-019	Subject/class report	4.6 / DS-Report-02	Report Module	TC-Report-02	P	—
SAM-F-020	Download/print report	4.6 / DS-Report-03	Report Module	TC-Report-03	P	—

Req ID	Requirement (Short)	Section Ref / Design Spec	Module	Test Case(s)	Status	Comments
SAM-NF-001	Response time ≤ 3 seconds	5 / DS-Perf-01	Web/Application Module	TC-Perf-01	P	Performance testing required
SAM-NF-002	99% availability	5 / DS-Reliability-01	System Module	TC-Reliability-01	P	—
SAM-NF-003	User-friendly interface	5 / DS-UI-01	User Interface	TC-UX-01	P	—
SAM-NF-004	Data protection	5 / DS-Security-01	Security Module	TC-Sec-01	P	—
SAM-NF-005	Database backup and recovery	5 / DS-Recovery-01	Database Module	TC-Recovery-01	P	—
SAM-NF-006	Maintainability	5 / DS-Maint-01	Application Module	TC-Maint-01	P	—
SAM-NF-007	Multiple-user support	5 / DS-Scale-01	Application Module	TC-Load-01	P	Load testing required
SAM-SEC-001	Authentication required	5.1.2 / DS-Security-01	Security Module	TC-SEC-01	P	—
SAM-SEC-002	Role-based authorization	5.1.2 / DS-Security-02	Security Module	TC-SEC-02	P	—
SAM-SEC-003	Secure password storage	5.1.2 / DS-Security-03	Authentication Module	TC-SEC-03	P	Passwords must be hashed
SAM-SEC-004	Protect student attendance data	5.1.2 / DS-Security-04	Access Control Module	TC-SEC-04	P	—
SAM-SEC-005	Audit important actions	5.1.2 / DS-Security-05	Audit Module	TC-SEC-05	P	Timestamp required

9. Conclusion

This document has specified the functional requirements (20, Section 4), non-functional requirements (7, Section 5), security objectives and requirements (2 and 5 respectively, Section 5.1), quality attributes and acceptance criteria (Section 6), use-case models (Section 7), and full requirements traceability (Section 8) for the Student Attendance Management System.

The team has committed to a Python/Django backend with a MySQL database (Section 2.1). The requirement set establishes a three-role access architecture (Student, Faculty, and Administrator) with attendance recording, calculation, and reporting as core functionalities. This SRS forms the baseline against which the subsequent project plan, test plan, and implementation deliverables will be developed and verified.

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10. Changes in Version 1.1

This version is updated to reflect the work done in Sprints 1 and 2 (19 to 30 October 2026). No requirement was added or removed. The changes refine wording that the SAD (v1.0, ADR-004) had already flagged, and record the test status.

- SAM-F-012 and SAM-F-013 (Section 4.4): attendance is stored per student, subject, date, and one-hour time slot, and duplicates are refused per time slot. A two-hour lab is two records, so it counts as two classes; lab and theory of one subject still count together in one percentage.
- UC-01 (Section 7.4): marking picks a date and one or more time slots and saves in one transaction; a saved session can be corrected with a reason (SAM-F-014).
- RTM (Section 8): all 32 requirements (20 functional, 7 non-functional, 5 security) have passing test cases; status changed from N to P.

Decisions made during implementation that fill in details the requirements left open (the requirements themselves are unchanged):

- SAM-F-014: only the faculty member assigned to the class and subject can modify its attendance, and a reason is required for every change.
- SAM-F-017: the threshold is set by the administrator and defaults to 75%; warnings appear on the student's dashboard and attendance page, and faculty see a list of their students below it.
- SAM-F-018 to SAM-F-020: reports take an optional date range; downloads are CSV and PDF, and the page prints without the menus.
- SAM-SEC-003: an account is locked for 15 minutes after 5 failed logins. SAM-NF-004: idle sessions expire after 30 minutes, and cookies are HTTPS-only in production.
- SAM-NF-005: backups are made and restored with two commands (backup_db, restore_db), each recorded in the audit log.

Where the implementation differs from the SAD or the STP, the difference and the reason are listed in docs/DEVIATIONS.md in the project repository.